

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Double-Blind, Placebo-Controlled Study of Atezolizumab Plus Carboplatin and Etoposide With or Without Tiragolumab in Patients With Untreated Extensive-Stage Small Cell Lung Cancer **Study Title:** Study of Atezolizumab Plus Carboplatin and Etoposide With or Without Tiragolumab in Participants With Untreated Extensive-Stage Small Cell Lung Cancer **Short Title/Acronym:** SKYSCRAPER-02C **Protocol Number:** Y042373 **NCT Number:** NCT04665856 **Posting ID:** 526169152205429311 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Tiragolumab (MTIG7192A) + Atezolizumab (Tecentriq) + Carboplatin + Etoposide **Phase of Development:** Phase III (PHASE3) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of tiragolumab plus atezolizumab combined with carboplatin and etoposide (CE) compared with placebo plus atezolizumab and CE in participants with untreated extensive-stage small cell lung cancer (ES-SCLC). **Secondary Objectives:** To evaluate the Objective Response Rate (ORR), Overall Survival (OS), and Progression-Free Survival (PFS) across analysis sets, along with the safety and tolerability of the investigational regimen.

2.2 Background & Rationale Small cell lung cancer (SCLC) is a highly malignant neoplastic process that is composed of small ovoid cells. Historically, SCLC was called oat cell carcinoma because the microscopic appearance of the cells was felt to resemble oats. It usually originates near the bronchi and typically grows and metastasizes quickly. While the standard first-line treatment (carboplatin or cisplatin plus etoposide) initially yields sensitive responses, most patients relapse within months. Tiragolumab is a novel monoclonal antibody targeting TIGIT, an immune checkpoint receptor. This study explores whether dual immune checkpoint blockade targeting both TIGIT (tiragolumab) and PD-L1 (atezolizumab), added to a standard chemotherapy backbone, can act synergistically to enhance the immune response and prolong survival in ES-SCLC patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional Control Type: Placebo-controlled Blinding: Double-blind Randomization: Randomized Criteria Type: PHASE_REQUIREMENT (Trial must be in PHASE3)

3.2 Planned Enrollment & Duration Target Enrollment: Approximately 123 participants Number of Sites: Multicenter (Study sites located in China) Estimated Study Start: February 2020 Estimated Primary Completion: September 2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 2. Histologically or cytologically confirmed Extensive-Stage Small Cell Lung Cancer (ES-SCLC) per the modified Veterans Administration Lung Study Group (VALG) staging system. 3. No prior systemic treatment for ES-SCLC. Participants who received prior chemoradiotherapy for limited-stage SCLC must have had treatment with curative intent and a treatment-free interval of at least 6 months. 4. Measurable diseases as defined by RECIST v1.1. 5. Submission of a pre-treatment tumor tissue sample. 6. Adequate hematologic and end-organ function. 7. Not receiving therapeutic anticoagulation (INR and aPTT $\leq 1.5 \times$ ULN) OR receiving therapeutic anticoagulation on a stable regimen. 8. Negative test results at screening for HIV, active Hepatitis B (HBsAg negative or specific requirements met for HBsAb/HBcAb/HBV DNA), Hepatitis C (HCV antibody negative or negative HCV RNA), and Epstein-Barr virus (EBV VCA IgM negative or negative EBV PCR). 9. Reproductive requirements: Women of childbearing potential and men must agree to remain abstinent or use contraceptive methods, and refrain from donating eggs/sperm.

4.2 Exclusion Criteria 1. Symptomatic or actively progressing central nervous system (CNS) metastases, spinal cord compression, or leptomeningeal disease. 2. Uncontrolled pleural effusion, pericardial effusion, ascites, or uncontrolled/symptomatic hypercalcemia. 3. Known clinically significant liver disease (active viral, alcoholic, or inherited liver disease, cirrhosis) or current alcohol abuse. 4. Active or history of autoimmune disease, immune deficiencies, or history of idiopathic pulmonary fibrosis, organizing pneumonia, drug-induced pneumonitis, or evidence of active pneumonitis. 5. Known active tuberculosis, severe chronic or active infection, or treatment with therapeutic oral/IV antibiotics. 6. Significant cardiovascular disease or a major surgical procedure other than for diagnosis. 7. Prior allogeneic bone marrow transplantation or solid organ transplant. 8. Prior treatment with CD137 agonists, T-cell co-stimulating, or immune checkpoint blockade therapies; treatment with systemic immunostimulatory or immunosuppressive agents. 9. Known hypersensitivity to Chinese Hamster Ovary (CHO) cell products,

tiragolumab/atezolizumab formulations, or allergic reactions to carboplatin or etoposide. 10. Administration of a live, attenuated vaccine. 11. Malignancies other than SCLC within 5 years prior to randomization. 12. Pregnancy or breastfeeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * *Induction:* Tiragolumab (600 mg) + Atezolizumab (1200 mg) + Carboplatin + Etoposide (ATC Code: L01CB01; Trade names: Etopophos, Vepesid, Toposar) administered on Day 1 of each 21-day cycle for 4 cycles. * *Maintenance:* Tiragolumab (600 mg) + Atezolizumab (1200 mg) administered every 3 weeks (Q3W) on Day 1 of each 21-day cycle. * *Comparator Arm:* Placebo + Atezolizumab + CE (induction) followed by Placebo + Atezolizumab (maintenance). **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** 21-day cycles, continued until disease progression, unacceptable toxicity, or loss of clinical benefit.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-related toxicities (e.g., anti-emetics) and stable anticoagulant regimens. **Prohibited:** Systemic immunosuppressive medications, systemic immunostimulatory agents, therapeutic oral or IV antibiotics at screening, anti-viral therapy for HBV/HCV, and live/attenuated vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Tissue Sample, Viral Screen, CT/MRI Scans, Baseline Labs
Baseline	Day 0	Randomization, First Dose (Cycle 1, Day 1)
Interim	Q3W (Every 3 Weeks)	Treatment administration, Safety Labs, AE Assessment
Tumor Assessment	Every 6 Weeks	CT/MRI Scans to assess response per RECIST v1.1
End of Study	Variable	End of Treatment Visit, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Investigator-Assessed Progression-Free Survival (PFS) in the Primary Analysis Set (PAS):** Time from randomization to the first occurrence of disease progression (PD) per RECIST v1.1 ($\geq 20\%$ increase in sum of diameters with absolute increase ≥ 5 mm) or death from any cause. * **Overall Survival (OS) in the PAS:** Time from the date of randomization to the date of death from any cause.

7.2 Secondary Endpoints * **PFS in the Full Analysis Set (FAS):** Progression-Free Survival assessed within the FAS. * **OS in the**

FAS: Overall Survival assessed within the FAS. * **Investigator-Assessed Confirmed Objective Response Rate (ORR) in the PAS:** Percentage of participants with a confirmed objective response.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via patient reporting and clinical/laboratory evaluations prior to each cycle. Immune-related AEs (irAEs) specifically monitored due to immune checkpoint blockade. **Serious Adverse Events (SAEs):** Reporting timeline typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to conduct periodic reviews of safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Primary Analysis Set (PAS) and Full Analysis Set (FAS) for primary and secondary efficacy outcomes; Safety Set for all subjects receiving at least one dose of study drug. **Sample Size Justification:** Powered to detect a statistically significant difference in PFS and OS between the Tiragolumab/Atezolizumab/CE arm and the Placebo/Atezolizumab/CE arm. **Handling of Missing Data:** Kaplan-Meier (K-M) method is used to estimate PFS and OS time-to-event rates. Right-censoring is applied for missing data or patients lost to follow-up.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Data captured via electronic Case Report Forms (eCRF) with standardized data clarification queries.

11. Study Identifiers & Governance

Primary ID: Y042373 **Registry Number:** NCT04665856 **Posting ID:** 526169152205429311 **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche) **Study Title:** SKYSCRAPER-02C **Official Regulatory Title:** A Phase III, Randomized, Double-Blind, Placebo-Controlled Study of Atezolizumab Plus Carboplatin and Etoposide With or Without Tiragolumab in Patients With Untreated Extensive-Stage Small Cell Lung Cancer

12. Clinical Framework (Oncology Specific)

Disease Area: Small Cell Lung Carcinoma / Small Cell Lung Cancer
Preferred Name: Small cell carcinoma of lung (UMLS: Small Cell Lung Carcinoma) **Semantic Type:** Neoplastic Process **Terminology Codes:** MeSH: D055752 | HPO: HP:0030357 | SNOMED: 254632001
Diagnosis Threshold: Histologically or cytologically confirmed ES-SCLC **Medical Methodology:** Immune Checkpoint Inhibitor Therapy + Platinum-based Chemotherapy Doublet **Optimization Target:** Synergistic targeting of PD-L1 and TIGIT pathways to prevent tumor immune evasion

13. Study Procedures & Methodology

Management Pathway: 1st Line (1L) Systemic Therapy for ES-SCLC
Education Protocol (HCP): Investigator training on protocol requirements, RECIST v1.1 criteria evaluation, and immune-mediated adverse event management. **Education Protocol (Patient):** Informed consent processing, AE self-reporting instructions, contraception compliance, and treatment scheduling. **Quality Audit Mechanism:** Central imaging review and consistent monitoring of safety data by the Sponsor and DMC.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until disease progression, death, or study withdrawal. **Onsite Visit Cadence:** Every 3 weeks (Q3W) corresponding with IV infusion cycles. **Remote Monitoring Frequency:** Interim safety check-ins as clinically indicated. **Checklist Review Interval:** Tumor imaging and RECIST v1.1 evaluation every 6 weeks.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				